Code No: 136EV **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, March - 2024
DATA STRUCTURES
(Common to EEE, ECE, CSE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

Used paper March-2024

PART – A

(25 Marks)

1.a) Define time complexity and space complexity. [2]

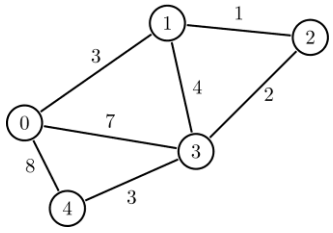
b) Represent the following sparse matrix as a linked list. [3]

$$A = \begin{bmatrix} 5 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 & 0 \end{bmatrix}$$

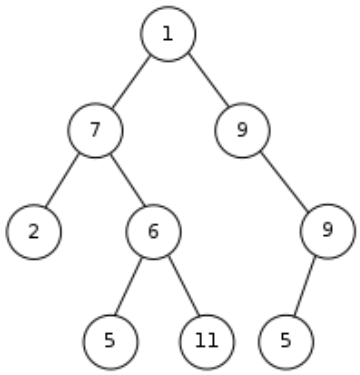
c) Write any four applications of stack. [2]

d) Find the value of the given postfix expression.
AB+C*DE-F/+G+
where A = 3, B = 2, C = 5, D = 4, E = 6, F = 9, G=12 [3]

e) Represent the following graph using adjacency matrix. [2]



f) Represent the following binary tree using array. [3]

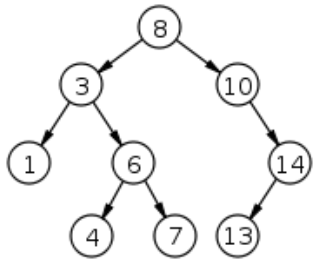


g) What is collision? How collisions are handled in hashing? [2]

h) How many comparisons are required using linear search and binary search to search the key 46 in the following elements? 12, 32, 34, 36, 40, 46, 50, 52, 53 [3]

i) Define Binary Search Tree and AVL Tree. [2]

j) Write in-order, pre-order, post-order traversal sequences for the following binary search tree. [3]



Used paper March-2024

PART – B

(50 Marks)

2.a) Write a function to insert an element at the end of the linked list.

b) Write an algorithm to insert an element at the front of circular linked list. [5+5]

OR

3.a) Write a function to add two sparse matrices.

b) Write a function to find GCD of two numbers using recursion. [5+5]

4.a) Write the program to implement operations of stack using arrays.

b) Write the program to convert given infix expression into postfix expression. [5+5]

OR

5.a) Write the program to implement operations of queue using linked list.

b) What is deque (Double ended queue)? Explain insert and delete operations of deque. [5+5]

6.a) Write a procedure to insert an element into max heap.

b) Define Graph. Write a function to traverse the graph using BFS. [5+5]

OR

7.a) Write a function to delete an element from binary tree.

b) Differentiate between DFS and BFS. [5+5]

8.a) Create a hash table using the following values.
98, 73, 21, 58, 88, 14, 37, 77
The hash function is h(x) =x%7.

b) Write a program to implement insertion sort. [5+5]

OR

9.a) Use selection sort to sort the following elements. Show the array elements at each step.
23, 21, 11, 43, 32, 55, 2, 9.

b) Write a program to search for an element using binary search. [5+5]

10.a) Write a function to insert an element into a binary search tree.

b) Explain various rotations of the AVL tree. [5+5]

OR

11.a) Write the process to insert and delete an element in B-Tree.

b) Write an algorithm for The Knuth-Morris-Pratt algorithm. [5+5]

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